Training-Dynamics-Coupled Fidelity for Machine-Centric Dataset Compression

A Methodology Draft for Resource-Aware Large-Scale Visual Learning

Anonymous Draft for Research Development

Abstract

Modern visual training pipelines repeatedly stream high-fidelity images across many epochs, even though a model does not require the same level of input detail at every stage of optimization. This creates a mismatch between *human-centric storage* and *machine-centric training needs*, especially when researchers are constrained by limited TPU/GPU quotas, I/O bandwidth, or storage. We propose **Training-Dynamics-Coupled Fidelity (TDCF)**, a new methodology for compressing training data according to what the model is *currently sensitive to* during optimization rather than what humans perceive as visually important. The key idea is to estimate an *instantaneous input sensitivity spectrum* from gradients in a warm-up phase, fit a monotone fidelity schedule over training time, and then serve each image at the coarsest fidelity level that still preserves the gradient-active information needed at that stage. Unlike static perceptual compression, the proposed method is explicitly machine-centric; unlike proxy-based importance scoring, it uses the target model's own gradients. The draft is designed as a *separate paper* from existing spatial state-space model work: it can be evaluated with SSM backbones such as CS-Mamba, but it does not depend on any specific characteristic-flow architecture. This document presents the problem formulation, related work, methodology, theoretical justification, and an experimental plan suitable for an ICLR-style submission.

1 Introduction

Deep models are usually trained by repeatedly reading the same dataset for tens or hundreds of epochs. In standard ImageNet-style training, this means that the system repeatedly spends bandwidth, storage, and accelerator time on full-fidelity images, even though optimization dynamics evolve dramatically across training. Early epochs often learn coarse category structure; later epochs refine class boundaries; very late epochs focus on hard distinctions. Yet the training pipeline usually treats every epoch as if the model always needs the same input fidelity.

This is a practical problem and a scientific one. Practically, researchers with limited hardware budgets are forced to choose between smaller models, shorter training runs, or reduced datasets. Scientifically, current data compression pipelines remain dominated by *human perception*: a codec discards what humans are less sensitive to, not what a classifier, segmenter, or state-space model finds irrelevant. For machine learning, those two notions need not coincide.

This paper starts from a simple question:

> *Can we compress training data according to the information that is relevant to the model at its current stage of learning, while preserving spatial structure and minimizing accuracy loss?*

We propose **Training-Dynamics-Coupled Fidelity (TDCF)**. The core idea is to measure, during a small pilot phase, how sensitive the model's loss is to different frequency bands and spatial residuals of the input. From these measurements we derive a *fidelity schedule*: early epochs consume coarse global structure, while later epochs progressively unlock finer details and harder local residuals. The resulting system is neither conventional image compression nor token pruning inside the model. It is an *offline dataset representation strategy* guided by the learning dynamics of the target model.

The proposed paper is intentionally separate from prior CS-Mamba methodology. That earlier direction focuses on preserving spatial structure *inside* the model's selective state-space processing. By contrast, the present work focuses on preserving machine-relevant information *before the model sees the input*, so that large-scale training becomes more storage- and bandwidth-efficient. The two directions are complementary, but their problem statements, experimental protocols, and claims are different enough to justify separate papers.

Main contributions. This draft makes four concrete contributions:

- It formulates **machine-centric dataset fidelity** as a dynamic quantity tied to optimization progress rather than a static property of the image.
- It introduces the **instantaneous input sensitivity spectrum**, a gradient-based object that measures which transformed input components the model currently uses.
- It proposes a **coarse-to-fine storage and loading pipeline** that preserves spatial layout while discarding gradient-inactive detail at each stage of training.

- It provides a **local theoretical bound** showing that the training loss perturbation due to fidelity truncation is controlled by omitted gradient mass and second-order curvature.

2 Problem Statement

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be a labeled image dataset with $x_i \in \mathbb{R}^{H \times W \times C}$ and $y_i \in \{1, \ldots, K\}$. Let f_θ be a visual model trained by minimizing cross-entropy loss

$$\mathcal{L}(\theta; x, y) = -\log p_\theta(y \mid x). \tag{1}$$

For a standard training run over E epochs, each image is read and decoded many times. If images are stored at a single high-fidelity format, total training I/O grows linearly with E, even though the information required by the optimizer may change substantially across epochs.

We seek a family of dataset representations

$$x_i^{(0)}, x_i^{(1)}, \ldots, x_i^{(L)}, \tag{2}$$

ordered from lowest to highest fidelity, and a schedule $a(e) \in \{0, \ldots, L\}$ for epoch e, such that training with $x_i^{(a(e))}$:

1. preserves the spatial information needed by the model,
2. reduces storage and/or data movement,
3. maintains top-1 accuracy within a small tolerable drop,
4. does not require a separate proxy model to define importance.

The central challenge is that fidelity should be defined *for the model*, not for humans. JPEG-like perceptual criteria are insufficient because edges or fine structures that appear visually minor may be decisive for classification. Conversely, some high-frequency detail may be nearly irrelevant to the target model during most of training. TDCF addresses this by learning fidelity from the target model's own gradient field.

3 Related Work

3.1 State-space models and efficient visual backbones

Structured state-space models (SSMs) have emerged as strong alternatives to quadratic self-attention for long-range sequence modeling. S4 showed that carefully parameterized state-space layers can model long dependencies efficiently [1]. Mamba later introduced *selective* state spaces, making the recurrence input-dependent while retaining linear-time sequence processing [2]. Vision-oriented Mamba variants extend these ideas to images, but they still raise important questions about ordering, spatial locality, and what visual information is actually necessary for learning [3].

The present paper does not modify the internals of an SSM. Instead, it targets the *data interface* between storage and model. This makes it complementary to model-side contributions such as CS-Mamba or other spatially aware SSM designs.

3.2 Human-centric versus machine-centric compression

Classical image compression prioritizes perceptual quality for humans. Recent machine-centric or task-aware work argues that compression should instead preserve features important to downstream tasks. The recent survey on human and machine coding highlights this shift clearly [4]. A related line studies task-driven image compression, where the representation is optimized for visual analytics or recognition rather than display quality. However, most existing work still uses a *static* target representation or optimizes a codec jointly with a fixed task.

Our setting is different in two ways. First, we focus on *training-time* data usage, not merely inference-time compressed sensing. Second, fidelity is *time-varying*: the same image may be served coarsely in early epochs and more finely later.

3.3 Token pruning, masking, and adaptive compute

Several methods reduce compute by pruning or merging tokens inside the model. DynamicViT prunes tokens adaptively during inference or training [5]; ToMe merges tokens to speed up ViTs [6]. Self-supervised masking methods such as MAE and BEiT also show that learning can proceed from partial or corrupted visual inputs [7, 8]. These methods are important evidence that not all raw input detail is necessary at all times.

Still, token pruning and masked pretraining operate *after loading the input*. They reduce internal computation, whereas TDCF aims to reduce the data served from storage in the first place. This is especially relevant when the bottleneck is storage or accelerator quota rather than model FLOPs alone.

3.4 Spectral bias and frequency-dependent learning dynamics

Neural networks are known to exhibit a spectral bias: they tend to fit low-frequency structure before high-frequency detail [9]. In vision, this suggests that training dynamics are not fidelity-invariant. If coarse information is learned first, then a static full-fidelity data stream is likely redundant during large portions of training. TDCF operationalizes this observation using gradients with respect to transformed input coefficients, yielding a measurable, model-specific fidelity schedule.

3.5 Feature disagreement across architectures

Different backbones can achieve similar accuracy while relying on different visual cues. Geirhos et al. showed that ImageNet-trained CNNs can be strongly texture-biased rather than shape-biased [10]. This is precisely why we avoid a proxy network in the proposed method. TDCF estimates what matters from the *same model being trained*, reducing the transferability gap that often weakens proxy-based compression schemes.

4 Proposed Method: Training-Dynamics-Coupled Fidelity

4.1 Overview

TDCF has two phases:

1. **Pilot measurement.** Train the target model for a short warm-up period on a small subset or a small number of epochs while measuring gradient sensitivity of transformed input components.
2. **Adaptive full training.** Store or serve each training image at fidelity levels chosen by a monotone schedule derived from the pilot statistics.

The method is model-agnostic at the training-pipeline level, but it is *model-specific in measurement*: the schedule is learned from the target model's own gradients.

4.2 Multi-fidelity dataset representation

Let $\mathcal{T}$ be a transform that separates coarse and fine information. In practice, $\mathcal{T}$ may be a block DCT, a wavelet transform, or a patchwise frequency decomposition. For an image x, write transformed coefficients as

$$c = \mathcal{T}(x), \qquad c = \{c_1, \dots, c_M\}. \tag{3}$$

We define a nested family of masks

$$\mathcal{M}^{(0)} \subseteq \mathcal{M}^{(1)} \subseteq \cdots \subseteq \mathcal{M}^{(L)} \subseteq \{1, \dots, M\}, \tag{4}$$

where lower indices retain fewer coefficients. The reconstructed fidelity-ℓ image is

$$x^{(\ell)} = \mathcal{T}^{-1}(\mathbf{1}_{\mathcal{M}^{(\ell)}} \odot c)\,. \tag{5}$$

Because the masks are nested, fidelity increases monotonically.

A purely global low-pass scheme is often too crude for vision tasks because it may destroy local details at semantically important locations. Therefore TDCF uses a **hybrid representation**:

$$x = x_{\text{base}} + r, \tag{6}$$

where x_{base} is a low-frequency reconstruction preserving the full image layout, and r is a collection of spatially localized high-frequency residuals over patches. This lets us retain the global geometry of the scene while selectively restoring local detail where training dynamics indicate it matters.

4.3 Instantaneous input sensitivity spectrum

For a mini-batch $\mathcal{B}_t$ at optimization step t, define the sensitivity of coefficient k as

$$\Phi_t(k) = \mathbb{E}_{(x,y)\sim\mathcal{B}_t}\left[\left\|\frac{\partial \mathcal{L}(\theta_t; x, y)}{\partial c_k}\right\|_1\right]. \tag{7}$$

The vector $\Phi_t \in \mathbb{R}^M$ is the **instantaneous input sensitivity spectrum**. Intuitively, it measures how much the current loss would change if transformed coefficient k were perturbed. Large values mean the model is currently using that component; small values mean that component is locally unimportant at step t.

To stabilize the estimate, we aggregate across a window of steps belonging to epoch e:

$$\bar{\Phi}_e(k) = \frac{1}{T_e} \sum_{t \in \text{epoch } e} \Phi_t(k). \tag{8}$$

We normalize this into a distribution over coefficients or frequency bands:

$$p_e(k) = \frac{\bar{\Phi}_e(k)}{\sum_{j=1}^{M} \bar{\Phi}_e(j) + \varepsilon}. \tag{9}$$

If coefficients are grouped into bands $b = 1, \dots, B$, we define band sensitivity

$$s_e(b) = \sum_{k \in \text{band } b} p_e(k). \tag{10}$$

4.4 Spatial residual sensitivity

Frequency alone is not enough. A classifier may need fine residuals only in specific patches. Let the image be partitioned into patches $p \in \{1, \dots, P\}$. Let r_p denote the high-frequency residual stored for patch p. We define a patch sensitivity score

$$g_e(p) = \mathbb{E}_{(x,y)}\left[\left\|\frac{\partial \mathcal{L}}{\partial r_p}\right\|_1\right]. \tag{11}$$

This yields a spatial importance map over the patch grid. We then keep:

- the low-frequency base for *all* patches, preserving the full spatial arrangement, and
- high-frequency residuals only for the top-q_e patches under $g_e(p)$ at epoch e.

Thus, TDCF is not merely a low-pass filter. It is a *spatially faithful coarse-to-fine representation.*

4.5 From sensitivity to a monotone fidelity schedule

Given band sensitivities $s_e(b)$, define cumulative active mass

$$S_e(m) = \sum_{b=1}^{m} s_e(b). \tag{12}$$

For a target retention ratio $\eta_f \in (0, 1)$, choose the smallest band cutoff

$$K_e = \min\{m : S_e(m) \geq \eta_f\}. \tag{13}$$

Similarly, for patch residuals, choose the smallest quota q_e such that the retained spatial residuals account for at least η_s of patch sensitivity mass:

$$\sum_{p \in \mathrm{Top}(q_e)} \tilde{g}_e(p) \geq \eta_s. \tag{14}$$

Here $\tilde{g}_e$ is the normalized patch sensitivity map.

Because the raw estimates are noisy, we fit monotone schedules $K(e)$ and $q(e)$ using isotonic regression or a monotone spline. The final training rule at epoch e is:

$$x_e^{\mathrm{serve}} = x^{\left(K(e), q(e)\right)}, \tag{15}$$

meaning that the data loader serves the base image with the first $K(e)$ active bands and the top-$q(e)$ spatial residuals.

4.6 Pilot phase and periodic refresh

The schedule can be estimated from a pilot subset $\mathcal{D}_{\mathrm{pilot}} \subset \mathcal{D}$, e.g., 1%–5% of the training set. This reduces measurement cost. The steps are:

1. initialize model parameters and run warm-up training on $\mathcal{D}_{\mathrm{pilot}}$,
2. compute $\bar{\Phi}_e$ and g_e across pilot epochs,
3. fit $K(e)$ and $q(e)$,
4. train on the full dataset using those schedules.

If the schedule drifts during long training, TDCF can refresh the statistics every R epochs on a small held-out pilot stream. This gives a continuum between a one-shot schedule and fully online adaptation.

4.7 Training objective

The main optimization objective remains unchanged:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\mathcal{L}\big(\theta; x_{e(x)}^{\mathrm{serve}}, y\big) \right]. \tag{16}$$

Thus TDCF is a *training data policy*, not a new classifier loss. It can be used with cross-entropy, label smoothing, mixup, or other standard training components.

4.8 Algorithmic summary

Input: dataset $\mathcal{D}$, target model f_θ, transform $\mathcal{T}$, fidelity levels $\{\mathcal{M}^{(\ell)}\}$, pilot ratio ρ, sensitivity thresholds (η_f, η_s).
Step 1: Build a multi-fidelity storage pyramid for each image: full low-frequency base plus optional patchwise residual buckets.
Step 2: Run pilot training on ρN examples and accumulate $\bar{\Phi}_e$ and g_e.
Step 3: Convert sensitivity statistics into monotone schedules $K(e)$ and $q(e)$.
Step 4: Train on full data while the loader serves only the bands and residual patches specified by $(K(e), q(e))$.
Step 5: Optionally refresh schedules every R epochs.

5 Theoretical Justification

The goal is not to claim an unrealistic exact theorem for all nonlinear models. Instead, we provide a *local and honest* justification based on the current loss landscape.

Let c denote full transformed coefficients and let $\tilde{c} = c + \delta$ denote a truncated or perturbed version produced by a lower-fidelity representation. Consider the loss at fixed parameters θ_t and label y. A first-order Taylor expansion gives

$$\mathcal{L}(\theta_t; \tilde{c}, y) - \mathcal{L}(\theta_t; c, y) \approx \langle \nabla_c \mathcal{L}(\theta_t; c, y), \delta \rangle. \tag{17}$$

This immediately yields the following local bound.

Proposition 1 (First-order fidelity bound)**.** *Assume $\mathcal{L}(\theta_t; c, y)$ is twice differentiable in a neighborhood of c, and its Hessian with respect to c has operator norm at most H_t in that neighborhood. Then for any perturbation δ,*

$$\begin{aligned} |\mathcal{L}(\theta_t; c+\delta, y) - \mathcal{L}(\theta_t; c, y)| \leq \|\nabla_c \mathcal{L}(\theta_t; c, y)\|_\infty \|\delta\|_1 \\ + \frac{H_t}{2} \|\delta\|_2^2. \end{aligned} \tag{18}$$

Equivalently, if the omitted coefficients are exactly those with small gradient mass, then the first-order loss perturbation is small up to a second-order remainder.

Proof. Apply Taylor's theorem with remainder at point c in direction δ and bound the linear term with H"older's inequality. □

This proposition is modest but important. It says that dropping transformed components with low current gradient sensitivity is justified *locally* because the induced loss error is controlled by omitted gradient mass. In other words, the gradient spectrum is not merely heuristic; it directly upper-bounds the first-order training disturbance.

A similar statement applies to patch residuals. Let $r = [r_1, \ldots, r_P]$ be the stacked residuals and let δ_r zero out low-sensitivity patches. Then

$$|\Delta\mathcal{L}| \leq \sum_{p=1}^{P} \left\| \frac{\partial \mathcal{L}}{\partial r_p} \right\|_1 \|\delta_{r,p}\|_\infty + O(\|\delta_r\|_2^2). \tag{19}$$

Therefore retaining residuals for the patches with largest gradient mass minimizes a first-order surrogate of the training error.

Remark 1. *This is deliberately weaker than claiming exact sufficiency or model-agnostic invariance. The contribution is a practical and mathematically grounded criterion for* training-time fidelity selection*, not a universal theorem about all classifiers.*

6 Why This is Distinct from the CS-Mamba Paper

The current TDCF paper should remain separate from the characteristic/spatial Mamba paper for three reasons:

1. **Different scientific question.** CS-Mamba asks how to preserve spatial structure *inside* an SSM-style architecture. TDCF asks how to preserve machine-relevant information *in the dataset stream.* These are different problem statements.

2. **Different unit of novelty.** CS-Mamba contributes an architectural mechanism. TDCF contributes a training-data representation policy and a gradient-derived fidelity criterion.

3. **Different evaluation.** A model paper is judged by architecture ablations and benchmark accuracy. A data-fidelity paper must additionally report bytes read, compression ratio, training throughput, and accuracy versus storage curves.

That said, CS-Mamba is still a very natural *target backbone* in the experiments. If TDCF improves training efficiency for SSM-based visual models in particular, the two papers can strengthen each other while remaining cleanly separated.

7 Experimental Protocol

7.1 Datasets and backbones

A staged evaluation is recommended:

- **Stage 1:** Tiny-ImageNet and ImageNet-100 for fast schedule prototyping.

- **Stage 2:** Full ImageNet-1K once the protocol is stable.

Recommended backbones:

- an SSM-based model such as CS-Mamba or a Vision-Mamba-style baseline,

- a standard CNN such as ConvNeXt-T or ResNet-50,

- a ViT baseline such as ViT-S/16.

This shows whether the method is broadly useful and whether the learned schedules differ across model families.

7.2 Baselines

The most relevant baselines are:

1. **Full-fidelity training** (standard pipeline),

2. **Static low-pass compression** with a fixed band cutoff,

3. **Perceptual codecs** (JPEG/WebP at matched storage budgets),

4. **Random coefficient dropping** or random residual dropping,

5. **Model-side token pruning only** without storage-aware compression.

7.3 Evaluation metrics

The paper should report:

- Top-1 and Top-5 accuracy,

- accuracy drop relative to full-fidelity training,

- dataset storage ratio,

- bytes read per epoch,

- end-to-end training wall-clock,

- images/sec and accelerator utilization,

- stability of the learned schedules $K(e)$ and $q(e)$.

7.4 Ablation studies

Essential ablations include:

- transform choice: DCT versus wavelets,

- pilot ratio: 0.5%, 1%, 5%,

- retention thresholds η_f and η_s,

- one-shot schedule versus periodic refresh,

- low-frequency-only versus low-frequency-plus-spatial-residuals,

- same method across SSM, CNN, and ViT backbones.

8 Expected Outcomes and Risk Assessment

The most realistic target claim is not "95% compression with no loss." A more credible and stronger story is:

> *By serving coarse global structure early and progressively unlocking gradient-active detail later, TDCF reduces storage/I/O cost substantially while keeping classification accuracy within a small, predictable margin of standard training.*

Expected outcomes that would already make the paper valuable include:

- a clear monotone evolution of frequency sensitivity over training,
- a measurable reduction in bytes read and wall-clock training,
- less than 1%–2% top-1 drop at moderate compression ratios, and
- evidence that different architectures learn different schedules.

The main risks are also clear:

- the learned schedule from a pilot subset may not transfer perfectly,
- some tasks may require high-frequency detail earlier than expected,
- storage savings may not fully translate to wall-clock gains if compute dominates I/O.

These are manageable risks, and they lead to meaningful ablations rather than fatal weaknesses.

9 Conclusion

This document proposes a new standalone paper centered on *machine-centric, training-dynamics-aware dataset fidelity*. The key claim is modest but strong: the amount and type of information a model needs from its training data is not static, and this can be estimated from the model's own gradient sensitivity rather than from human perception or a separate proxy network. TDCF turns that observation into a concrete pipeline: build a multi-fidelity dataset, measure gradient-active bands and residual patches in a pilot phase, fit a monotone fidelity schedule, and train at the coarsest representation sufficient for the current stage of optimization.

As a research direction, this is cleanly separable from spatial SSM architecture work such as CS-Mamba. As a practical method, it directly addresses the reality of limited training budgets, bandwidth, and storage. That combination makes it a plausible and original candidate for a separate ICLR-style submission.

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